

$$\begin{aligned}
\dot{a}_{-1} &= - \left(\frac{\kappa_{-1}}{\kappa_0} + i\Delta_{-1} \right) a_{-1} + i \left(f_{-1}^{-1,-1,-1} |a_{-1}|^2 + 2f_{-1}^{0,0,-1} |a_0|^2 + 2f_{-1}^{1,1,-1} |a_1|^2 \right) a_{-1} + i f_{-1}^{0,1,0} a_0^2 a_1^* + s_{-1} \\
\dot{a}_0 &= - (1 + i\Delta_0) a_0 + i \left(2f_0^{-1,-1,0} |a_{-1}|^2 + f_0^{0,0,0} |a_0|^2 + 2f_0^{1,1,0} |a_1|^2 \right) a_0 + 2i f_0^{-1,0,1} a_{-1} a_0^* a_1 \\
\dot{a}_1 &= - \left(\frac{\kappa_1}{\kappa_0} + i\Delta_1 \right) a_1 + i \left(2f_1^{-1,-1,1} |a_{-1}|^2 + 2f_1^{0,0,1} |a_0|^2 + f_1^{1,1,1} |a_1|^2 \right) a_1 + i f_1^{0,-1,0} a_0^2 a_{-1}^* + s_1 \\
\frac{da_0}{d\tau} &= - (1 + i\Delta_0) a_0 + i \left[2 \left(f_0^{-1,-1,0} + f_0^{1,1,0} \right) \|a_p\|^2 \right] a_0 + 2i f_0^{-1,0,1} a_p^2 a_0^* \tag{1}
\end{aligned}$$